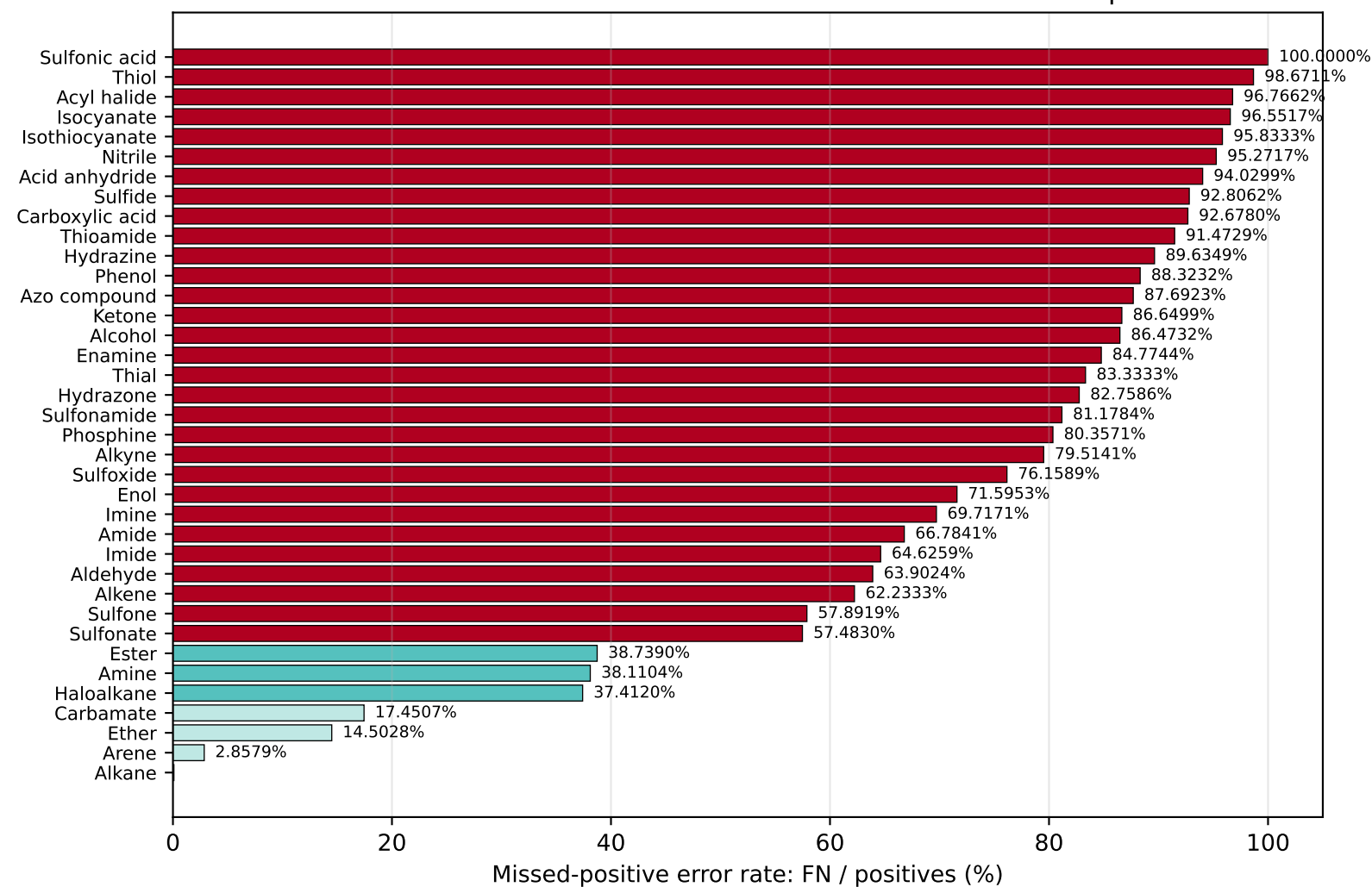
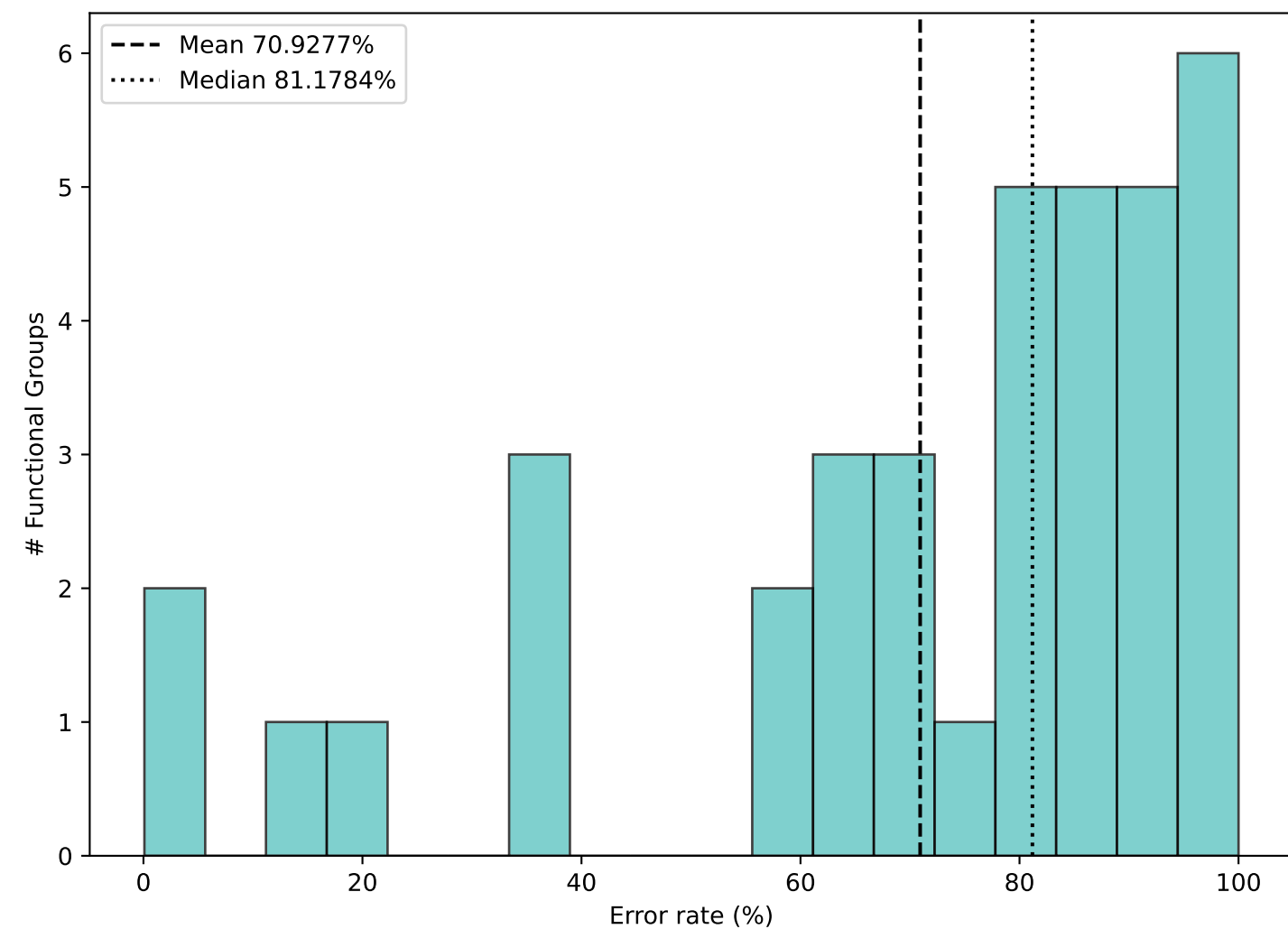


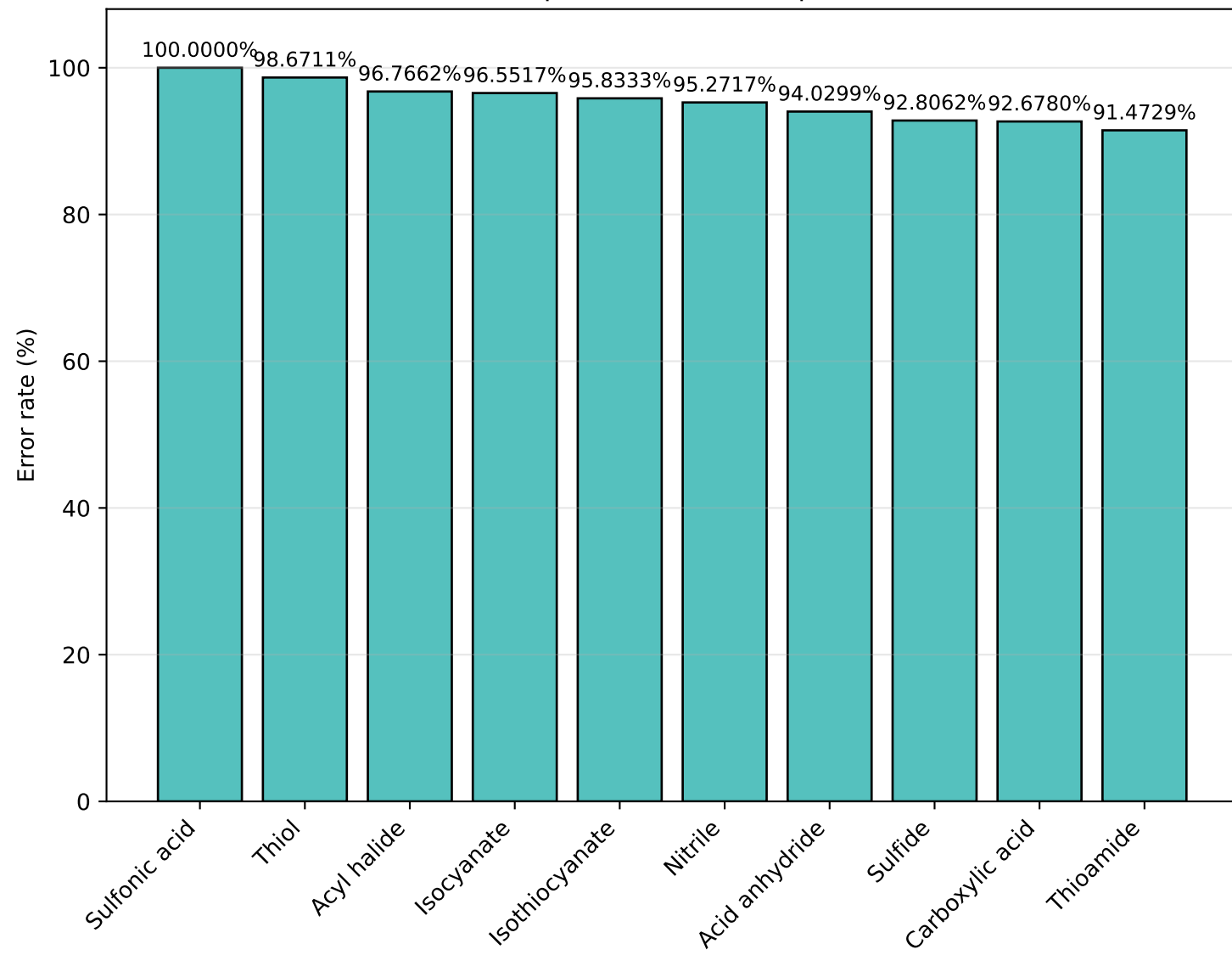
XGBoost H-NMR Validation Set - All 37 Functional Groups



Error-Rate Distribution



Top 10 Problem Groups



XGB00ST H-NMR VALIDATION SET ERROR ANALYSIS

Global metrics:

F1 micro: 0.7634
F1 macro: 0.3791
Precision micro: 0.8871
Recall micro: 0.6699
Hamming accuracy: 0.9464

Error-rate statistics:

Mean: 70.9277%
Median: 81.1784%
Std: 27.7533%
Max: 100.0000% (Sulfonic acid)
Min: 0.0889% (Alkane)

Problem classes:

Severe (>50%): 30 groups
High (20-50%): 3 groups
Moderate (5-20%): 2 groups
Low (<=5%): 2 groups